Course Outline Template

Indian Institute of Management Calcutta

Programme : MBA

Course Code :

Name of the Course: ……Generative AI in Action…..

Core / Elective: ……. Elective…… **Cap (for Elective)**……60…….

Credit: …1.5….

Academic Year & Term: …2026-27, Term VI (Post-mid)

Course Coordinator: Prof. Soumyakanti Chakraborty

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Introduction to Course /Course Description

Generative Artificial Intelligence is rapidly reshaping how organizations create value, redesign work, interact with customers, build knowledge assets, and make decisions. For managers, the challenge is no longer merely to understand what Generative AI tools can do, but to identify where they can create meaningful business value, how they can be embedded into organizational workflows, and how their risks can be governed responsibly.

This course takes a managerial and applied view of Generative AI. It introduces students to the foundations of Generative AI and Large Language Models, but focuses primarily on their business implications: the Generative AI value chain, data readiness, Retrieval-Augmented Generation, adoption challenges, productivity gains, workflow redesign, digital twins, agentic AI, evaluation, and governance. The course will help students move beyond casual tool usage and develop the ability to think like managers who can evaluate, design, and lead Generative AI initiatives.

Through examples, demonstrations, classroom discussions, and hands-on exposure to AI-enabled workflows, students will examine how Generative AI can support productivity, decision-making, innovation, and organizational change.

The course is designed for MBA students who wish to understand not only how Generative AI works, but also how it can be deployed thoughtfully, scaled responsibly, and aligned with business objectives.

Course Objective and Key-takeaways from the course

The course enables students to:

- Understand how Generative AI can create enterprise value through improved productivity, decision-making, knowledge work, and customer-facing applications.
- Assess Generative AI use cases by examining business relevance, data readiness, organizational readiness, scalability, and risk.
- Design AI-enabled and agentic workflows that combine Large Language Models, enterprise data, tools, human oversight, evaluation, and accountability.
- Think critically about the governance, responsible use, and future-of-work implications of Generative AI and agentic AI in organizations.

Programme Level Learning Goals (7)

Learning Goals **

1. Possess the ability to think critically and find solutions to managerial problems

    Learning Objective**: Learn to use analytical tools of various disciplines and functions in practical situations**

7. Acquaint themselves with established and emerging technologies for effective decision making

    Learning Objective:

    a) Familiarize themselves with different state-of-the-art computing environments and understand their benefits and limitations

    b) Be competent in the application of technology based concepts in modeling and solving managerial problems involving decision making

International Components in your course (if any)

None

Pre-requisites

None

Required Text Book(s):

Recommended Text Book(s)

Mollick, E. (2024). Co-intelligence: Living and working with AI. Portfolio/Penguin.

Iansiti, M., & Lakhani, K. R. (2020). Competing in the age of AI: Strategy and leadership when algorithms and networks run the world. Harvard Business Review Press.

Daugherty, P. R., & Wilson, H. J. (2024). Human + machine: Reimagining work in the age of AI. Harvard Business Review Press.

Farri, E., & Rosani, G. (2025). HBR guide to generative AI for managers. Harvard Business Review Press.

Course Pack Distribution to students

A collection of relevant articles/papers may be provided to the students during the course.

Technology enabled learning component for your course

Students will be expected to use Google Colab for the course:
https://research.google.com/colaboratory/

Colab will allow students to write and execute arbitrary python code through a browser, and is especially well suited to machine learning and data analysis. It is a hosted Jupyter notebook service that requires no setup to use, while providing free access to computing resources including GPUs.

Session Plan

S No.	Topic	Intended learning outcome	References
1	The Foundations of Generative AI	Understand the core idea behind language generation.	Mollick, E. Co-Intelligence: Living and Working with AI, Chapters. 1 and 2 Farri, E. and Rosani, G., "How AI Can Help Managers Think Through Problems," HBR.
2	Generative AI Fundamentals	Understand foundation models, instruction tuning, prompting, and the distinction between models, applications, and copilots.	Mollick, E. Co-Intelligence: Living and Working with AI, Ch. 3 Farri and Rossani. HBR Guide to Generative AI for Managers. Chapters 1 and 2
3	AI and Data Value Chains	Map how data, models, applications, workflows, and governance create enterprise value.	Iansiti, M. and Lakhani, K. R. Competing in the Age of AI, Ch. 3
4	Retrieval-Augmented Generation	Understand how enterprise knowledge can be connected to Large Language Models and what quality, cost, and governance issues arise.	
5	AI Adoption and the Productivity J-Curve	Analyze why AI value takes time and what organizational changes are needed for scaling.	Iansiti, M. and Lakhani, K. R. Competing in the Age of AI, Ch. 4 and Ch. 5 Burnham, K., "The Productivity Paradox of AI Adoption in Manufacturing Firms," MIT Sloan Ideas Made to Matter
6	Digital Twins	Examine digital twins as AI-enabled systems for simulation, scenario planning, and decision support.	Daugherty, P. R. and Wilson, H. J. Human + Machine, updated edition, Ch. 2 Kenny, G. and Pogrebna, G., "Digital Twins Can Help You Make Better Strategic Decisions," HBR.
7	Agentic AI Foundations	Understand agents, tool use, planning, multi-agent coordination, and managerial use cases.	Mollick, E. Co-Intelligence: Living and Working with AI, Ch. 6 What Is Agentic AI, and How Will It Change Work? HBR Stackpole, B., "Agentic AI, Explained," MIT Sloan Ideas Made to Matter.

8	Agentic Workflow Design	Design an enterprise workflow using agents, tools, approvals, escalation, and audit trails.	Daugherty, P. R. and Wilson, H. J. Human + Machine, updated edition, Ch. 8 Mantia, L., Chatterjee, S. and Lee, V. S., "Designing a Successful Agentic AI System," HBR.
9	AI Evals and Responsible Governance	Assess outputs, identify harms, and propose governance controls for responsible deployment.	Iansiti, M. and Lakhani, K. R. Competing in the Age of AI, Ch. 8 Stackpole, B., "Where to Look for Generative AI Risks," MIT Sloan Ideas Made to Matter Bojinov, I., "Keep Your AI Projects on Track," HBR.
10	Future of AI and Work	Reflect on Artificial General Intelligence, future skills, organizational redesign, and the changing nature of managerial work.	Mollick, E. Co-Intelligence: Living and Working with AI, Ch. 9: Daugherty, P. R. and Wilson, H. J. Human + Machine, Ch. 9 Autor, D., Mindell, D. A. and Reynolds, E. B., "Why the Future of AI Is the Future of Work," MIT Sloan Ideas Made to Matter

Evaluation Components

✓ Components	✓ Weightage
Assignments (Group)	**60 (4 x 15)**
End Term	**40**